

IC 141: Product Realization Technology

Course Code	IC 141
Course Name	Product Realization Technology
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	UG
Mutual Exclusion	Courses with high similarity not allowed to credit by the students after or along with this course
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum; 46th BoA (May 2022); 37th Senate (Oct 2022)
Revised	47th BoA (Sep 2022) and 37th Senate (Oct 2022) - renamed from "Product Realization Theory"
Recoded to	ME 212 by the 64th BoA (May 2026)
Discontinued	64th BoA (May 2026)

Course Contents

- **Introduction:** Engineering materials, their manufacturability and application (2 Hours)
- **PRT for Structural Applications (Metals & Alloys):** Casting (sand casting, permanent mold casting, investment casting), forming (Rolling, Forging, Extrusion, Sheet metal operations), machining (drilling, lathe, milling), joining, Additive manufacturing (SLS, SLM) (8 Hours)
- **PRT for Light weight applications (Polymers):** Classifications of plastics, blow molding, injection molding, extrusion, compression molding, vacuum forming, Additive manufacturing (FDM, SLM), laser machining, joining methods. (6 Hours)
- **PRT for High Temperature Applications (ceramics and glasses):** Powder manufacturing, mixing and blending, compacting, Sintering (with SPS), hot iso-static pressing (5 Hours)
- **PRT for High Performance Applications (Composites):** Microwave curing, compression molding, resin infusion microwave curing, pultrusion, filament winding (5 Hours)
- **PRT for Electronic Applications (Semiconductors):** Thin film deposition, photolithography, wet bulk micromachining/etching, screen printing, 3D printing, Fabrication of PCBs. (2 Hours)

Laboratory Modules

Sl. No.	Equipment	Experiment	Turns
1	Lathe, milling, shaper machine	Facing, turning and grooving operations on mild steel rod	1
2	Sintering	To demonstrate sintering process for ceramic/metallic powders	1
3	Microwave composite processing	To demonstrate manufacturing of composite laminates	1
4	Fitting	To make fillet, chamfer, drilling and tapping on mild steel flat	1
5	Welding	To perform arc welding, gas welding and spot welding and FSW	1
6	Sheet metal	To perform shearing, bending and riveting of galvanised iron sheet	1
7	Foundry	To sand cast an aluminium rod	1
8	CNC lathe and milling	To machine objects using CNC machining processes	1
9	Laser machining	To demonstrate laser machining of an acrylic sheet	1
10	Advanced Manufacturing	To demonstrate thin film deposition, lithography, screen printing, fabrication of PCBs.	1
11	Additive manufacturing	To create objects using advanced 3D printing processes	1
12	Injection molding	To understand the plastic injection molding process	1
			12

Text Books

1. Groover, M.P., **Fundamentals of modern manufacturing: materials, processes, and systems**, John Wiley & Sons, 2020.
2. Kalpakjian, S. and Schmid, S.R., **Manufacturing engineering and technology**, Prentice Hall, 2001.

References

None